

Minutes of Meeting (MoM)

Meeting Details

Meeting Title: FOTA Architecture Requirements Discussion

Date: 20-Aug-2026

Time: 03:00 PM to 04:00 PM

Organizer: [YASH SUNIL GIRAMKAR](#)

Attendees:

- Aniket Bhattacharya, Suresh Venkatesan, Manish Chandra, Vishal Shetye, Shivamkumar Chudasama, Amol Kulkarni, Shubham Jadhav, Amey Panse, Charu Chandan, Yogesh Aher

Objective: Review FOTA modem and cloud architecture requirements, clarify open queries, align on implementation approach, and finalize pending requirements.

Discussion Summary

1.Trigger & Scheduling

Points Discussed

- RTC trigger frequency.
- Dependency on the notify-next MQTT topic.

Decisions Taken

- If the last \$next/get request was sent 6 hours or more ago, a new \$next/get request should be published.
- The \$next/get request will be published only when the Ignition is ON.
- Heartbeat messages will also be generated only when the Ignition is ON, with a minimum frequency of once every 2 hours, subject to the Ignition ON condition.
- Every MQTT request must be retried at least three times before being considered failed.
- The notify-next topic will be completely removed.
- No MQTT topics will be subscribed to by default. MQTT publish/subscribe topics will open a session and establish a connection only when triggered by a relevant event.

2. Cloud Polling & Budget

Points Discussed

- Maximum number of polling requests allowed per day to the MQTT server.

Decision Taken

- A maximum of 4 polling events per day will be allowed.
- The polling count will reset at midnight.
- The device will persist the last polling count and the RTC time of the last polling request.
- Maintaining a persistent session for the notify-next topic may result in additional communication costs; therefore, the topic will be removed.

3. Job Document Lifecycle

Points Discussed

- Job lifecycle: When the job document should be fetched and the duration for which it should be persisted.
- Malformed job document handling: Expected system behavior and recovery strategy when an invalid or malformed job document is received.
- Supported operation types: The different types of operations supported by the system.

Decision Taken

- Every newly discovered job will require a fresh job document.
- If acquisition of a fresh job document fails after exhausting the configured cloud retry attempts, the current job context will be discarded, and the system will wait for the next RTC/Notify-Next event.
- If a job document is malformed or cannot be parsed, it will be discarded immediately, and a fresh job document will be requested according to the cloud retry policy.
- A valid JSON job document that fails FOTA validation will be treated as a recoverable document failure. A fresh job document will be requested using the cloud retry policy.
- Each validation retry will initiate a completely fresh job-document request. The same invalid document must not be repeatedly validated.
- The job document and relevant parsed metadata will be persisted only after successful parsing and complete FOTA validation.
- Any missing mandatory job-level or package-level field will result in whole-job pre-validation failure.
- Currently, only the fota operation is supported. Any unsupported operation type will result in job failure.

4. Version, Eligibility and Package Processing.

Points Discussed

- Job document and package version validation.
- Handling scenarios where the FOTA update can be skipped because the vehicle or ECU is already running the required version.

Decision Taken

- The FOTA Manager is responsible for validating package versions.
- The Manifest Manager runs at power-on, retrieves the current vehicle/ECU versions, and maintains a copy of this information in the filesystem. The FOTA Manager retrieves the current version information through the Manifest Manager API.
- The current ECU/vehicle version will be retrieved separately for each package immediately before evaluating that package.
- The Manifest Manager guarantees that the required current version information is available when queried by the FOTA Manager.
- An update is required when the following condition is met:
 - $\text{min_ver} \leq \text{current} < \text{target_ver}$
- Packages that are already at the target version are considered up to date and will be skipped.
- Version strings are alphanumeric identifiers (for example, GL437650201), and lexical comparison will be used for version evaluation.
- min_ver must be strictly less than target_ver . If this condition is not met, the job will be considered invalid.
- For an up-to-date package, the installation step will be bypassed. After sending the SKIPPED status and receiving the corresponding cloud acknowledgement, processing will proceed to the next package.
- Initially, min_ver will be reported by the cloud as 0. The value will be populated later, and the handling strategy for this transition remains TBD.
- seq_id will strictly determine the package processing order and govern the complete package lifecycle, including any future deployment or flashing activities. Any invalid package sequencing will result in job failure.

5. Download, Resume and Verification

Points Discussed

- Download, resume, and persistence policies for the device.
- Verification reporting and handling procedures.

Decision Taken

- Resume the download from the last successfully downloaded byte when HTTP supports ranged/resumable downloads.
- Current read, write, and download pointers may be stored in a separate filesystem file, resume_metadata.
- The resume metadata will contain at least:
 - package_id
 - downloaded_bytes
 - expected_package_size
- Any additional HTTP information required to support download resumption.
- HTTP download progress will be reported at every 25% checkpoint.
- Download resume information will be persisted at every 25% progress checkpoint to support recovery in case of an abrupt shutdown. In the event of a graceful shutdown, the download may resume from the exact point at which it was interrupted.
- If the resume metadata and the actual downloaded file are inconsistent, both current.fota and the associated resume metadata will be deleted, and the download will restart from byte zero.
- The hash will be calculated over the complete downloaded current.fota file and compared against the hash provided in the job document.
- Upon successful verification, the package status will be reported as VERIFICATION_COMPLETED. The system will not wait for cloud acknowledgement before proceeding with installation or downloading the next package.
- In the event of a hash failure requiring a re-download, the DOWNLOADING state must be persisted before deleting or resetting the package.

Key Decisions

- The system will not wait for cloud acknowledgement after receiving the job document. It is acceptable to proceed with the download and installation process even if an acknowledgement from the MQTT server has not been received.
- A malformed or invalid Manifest will result in a state where operations such as FOTA and OA cannot be reliably performed on the vehicle. In this condition, the FOTA state machine and OA operations will not resume normal operation and will remain in a waiting state until the next power cycle.
- After a power cycle, the system will attempt to resume normal FOTA and OA operations only after:

Successfully retrieving the versions of all vehicle ECUs, and

Verifying the overall integrity of the vehicle.

- Once these checks are successfully completed, normal FOTA and OA operations may resume.

Purpose: Review open TBD items and confirm final architecture decisions.

Prepared By: [YASH SUNIL GIRAMKAR](#)